

PN Junction Physics: Electrostatic Analysis and Current-Voltage Characteristics

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Abstract

This report presents a comprehensive computational analysis of PN junction physics, covering the formation of the depletion region, built-in potential, electrostatic field profiles, current-voltage characteristics governed by the Shockley diode equation, junction capacitance (both depletion and diffusion), and breakdown mechanisms. We analyze silicon PN junctions with varying doping concentrations, compute key parameters including built-in voltage ($V_{bi} = 0.7\text{--}0.9$ V), saturation current ($I_s \sim 10^{-15}$ A), and breakdown voltage ($V_{BR} \sim 10\text{--}100$ V), and examine the transition from thermal equilibrium to forward/reverse bias conditions.

1 Introduction

The PN junction forms the foundation of modern semiconductor devices, from diodes and bipolar junction transistors to solar cells and light-emitting diodes. When P-type (acceptor-doped) and N-type (donor-doped) semiconductors are brought into contact, carrier diffusion establishes a depletion region with an internal electric field that balances diffusion and drift currents at equilibrium.

Definition 1.1 (PN Junction) *A PN junction is a metallurgical junction between P-type semiconductor (excess holes, N_A acceptors/cm³) and N-type semiconductor (excess electrons, N_D donors/cm³), characterized by a built-in potential V_{bi} and depletion width W at thermal equilibrium.*

2 Theoretical Framework

2.1 Built-in Potential

At thermal equilibrium, the Fermi level is constant across the junction. The built-in potential barrier arises from the difference in quasi-Fermi levels before contact.

Theorem 2.1 (Built-in Voltage) *For a silicon PN junction at temperature T with intrinsic carrier concentration n_i :*

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) \quad (1)$$

where $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, $q = 1.6 \times 10^{-19}$ C is the elementary charge, and $n_i = 1.5 \times 10^{10}$ cm⁻³ for silicon at 300 K.

2.2 Depletion Region Analysis

Definition 2.1 (Depletion Width) Under reverse bias V_R or zero bias, the depletion region extends into both P and N sides:

$$W = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} + V_R)} \quad (2)$$

where $\epsilon_s = 11.7\epsilon_0 = 1.04 \times 10^{-12}$ F/cm is the silicon permittivity.

The electric field profile is triangular, peaking at the metallurgical junction:

$$E_{max} = \frac{qN_A x_p}{\epsilon_s} = \frac{qN_D x_n}{\epsilon_s} \quad (3)$$

2.3 Current-Voltage Characteristics

Theorem 2.2 (Shockley Diode Equation) The current through an ideal PN junction under applied voltage V is:

$$I = I_s \left[\exp \left(\frac{qV}{nkT} \right) - 1 \right] \quad (4)$$

where I_s is the saturation current (reverse-bias leakage) and n is the ideality factor ($n = 1$ for ideal diode, $n = 1.5-2$ for real diodes with recombination).

The saturation current depends on minority carrier properties:

$$I_s = qA \left(\frac{D_n n_{p0}}{L_n} + \frac{D_p p_{n0}}{L_p} \right) \quad (5)$$

where $D_{n,p}$ are diffusion coefficients, $L_{n,p}$ are diffusion lengths, and n_{p0}, p_{n0} are equilibrium minority carrier concentrations.

2.4 Junction Capacitance

Definition 2.2 (Depletion Capacitance) The voltage-dependent depletion capacitance per unit area:

$$C_j = \frac{\epsilon_s}{W} = \sqrt{\frac{q\epsilon_s}{2(V_{bi} + V_R)} \left(\frac{N_A N_D}{N_A + N_D} \right)} \quad (6)$$

Definition 2.3 (Diffusion Capacitance) Under forward bias, minority carrier storage contributes:

$$C_d = \frac{qI}{nkT} \tau_t \quad (7)$$

where τ_t is the transit time (minority carrier lifetime).

2.5 Breakdown Mechanisms

Remark 2.1 (Breakdown Voltage) At high reverse bias, two mechanisms cause breakdown:

- **Avalanche breakdown:** Impact ionization when $E_{max} \approx 10^6$ V/cm (larger V_{BR} for lightly doped junctions)
- **Zener breakdown:** Band-to-band tunneling for heavily doped junctions ($V_{BR} < 5$ V)

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Figure 1: Comprehensive PN junction analysis for silicon with $N_A = 10^{16} \text{ cm}^{-3}$ and $N_D = 10^{15} \text{ cm}^{-3}$: (a) Energy band diagram at thermal equilibrium showing built-in potential barrier of ?? V; (b) Triangular electric field profile with peak field ?? V/cm at the metallurgical junction; (c) Square-root dependence of depletion width on reverse bias voltage; (d) Current-voltage characteristics on linear scale showing exponential forward conduction and saturation reverse current ?? A; (e) Semi-logarithmic I-V plot comparing ideal ($n = 1$) and non-ideal ($n = 1.5$) diode behavior; (f) Voltage-dependent junction capacitance decreasing with reverse bias; (g) Linear relationship in $1/C^2$ vs V plot enabling extraction of built-in voltage; (h) Electric field expansion under increasing reverse bias; (i) Abrupt doping concentration profile with P-side acceptor density and N-side donor density.

3 Computational Analysis

4 Results

4.1 Junction Parameters at Thermal Equilibrium

begin{table}[H]
centering
caption{Computed Results}
begin{tabular}{lc@}
toprule Parameter & Value

midrule Parameter A & 3.14

Parameter B & 2.71

Parameter C & 1.41

bottomrule
end{tabular}
end{table}

4.2 Current at Different Voltages

4.3 Depletion Width vs Reverse Bias

5 Discussion

Example 5.1 (Asymmetric Depletion) *For this junction with $N_A/N_D = 10$, the depletion region extends approximately 10 times farther into the lightly-doped N-side than the P-side. This is evident from:*

$$\frac{x_n}{x_p} = \frac{N_A}{N_D} = 10$$

The asymmetry is critical for breakdown voltage design in power diodes.

Remark 5.1 (Forward Voltage Drop) *The forward voltage required to achieve significant current ($I \gg I_s$) is approximately: This ~ 0.7 V threshold is characteristic of silicon diodes and is widely used in circuit design.*

Example 5.2 (Breakdown Voltage Estimation) *The critical field for avalanche breakdown in silicon is approximately $E_{crit} \approx 3 \times 10^5$ V/cm. Using the maximum field relation: Higher breakdown voltages require lower doping concentrations.*

Remark 5.2 (Temperature Dependence) *The built-in voltage decreases with temperature at approximately -2 mV/K due to the T dependence in the logarithm and the exponential increase of $n_i^2 \propto T^3 \exp(-E_g/kT)$.*

6 Conclusions

This computational analysis demonstrates the fundamental physics of PN junctions:

1. The built-in potential for this silicon junction is $V_{bi} = ??$ V, determined by the doping ratio and intrinsic carrier concentration.
2. The depletion width at equilibrium is $W_0 = ??$ μm , expanding to $??$ μm under -5 V reverse bias.
3. The saturation current is $I_s = ??$ A, setting the leakage current floor in reverse bias.
4. Forward bias reduces the barrier, enabling exponential current growth with voltage according to the Shockley equation.
5. The junction capacitance decreases from $??$ pF at zero bias, following $C_j \propto (V_{bi} + V_R)^{-1/2}$ under reverse bias.
6. The estimated breakdown voltage for this doping level is approximately $??$ V, suitable for low-voltage rectification applications.

The computational framework enables rapid exploration of junction design parameters, essential for optimizing diodes, solar cells, and bipolar transistors.

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